

Hence, the correct option is (d).

Assessment Test II

- Thermosphere exists in plasma state.
Hence, the correct option is (d).
- $1 \text{ atm} \rightarrow 76 \text{ cm}$
 $0.05 \text{ atm} ?$
 $= 76 \times 0.05 \text{ cm} = 3.8 \text{ cm}$
Hence, the correct option is (a).
- The quantity of water vapour in the atmosphere differs from place to place and with change of season.
Hence, the correct option is (c).
- Combustion consumes but does not produce O_2 .
Hence, the correct option is (d).
- Respiration is a slow combustion.
Hence, the correct option is (c).
- Carbon particle burns with yellow flame in the middle zone of the candle flame.
Hence, the correct option is (b).
- A protective layer of Al_2O_3 is formed on Al upon reaction with O_2 .
Hence, the correct option is (b).
- When ammonia is burnt in limited supply of O_2 , N_2 and H_2O are produced.
Hence, the correct option is (d).
- $\text{FeS}_2 + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$
Hence, the correct option is (d).
- Tin-coated iron containers are used for storage of edible food stuffs.
Hence, the correct option is (c).
- Combustion is an exothermic reaction and rapid combustion produces a lot of heat. Hence oxyacetylene flame is used for welding purpose.
Hence, the correct option is (a).

- When air is passed through lime water, it turns milky because CO_2 is present in air and it retains its property.
Hence, the correct option is (a).

- The required order is CDAB.

- (C) SO_2
(D) Ozone
(A) Fly ash
(B) CFCs

Hence, the correct option is (c).

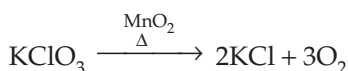
- The required order is BADC.

- (B) CuO
(A) Al_2O_3
(D) Fe_3O_4
(C) CaO

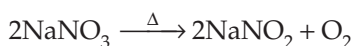
Hence, the correct option is (a).

- (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)

$\text{H}_2\text{O}_2 \rightarrow$ Heating is not required.



MnO_2 reduces the melting point as well as decomposition temperature of KClO_3 .



Hence, the correct option is (c).

Assessment Test III

- With increase in altitude, atmospheric pressure decreases; hence, the height of the mercury column decreases in barometer.
Hence, the correct option is (a).
- MnO_2 acts as a positive catalyst and decreases the melting point of KClO_3 during the preparation of O_2 from KClO_3 .
Hence, the correct option is (a).
- The correct order is DABC.
Hence, the correct option is (b).

4. $\text{Na} + \text{O}_2 \rightarrow \text{Na}_2\text{O}$; Number of oxygen atoms in one molecule is 1.

$\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_3\text{O}_4$; Number of oxygen atoms in one molecule is 4.

$\text{Al} + \text{O}_2 + \text{Al}_2\text{O}_3$; Number of oxygen atoms in one molecule is 3.

$\text{S} + \text{O}_2 \rightarrow \text{SO}_2$; Number of oxygen atoms in one molecule is 2.

Hence, the correct option is (b).

5. (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (A)

Hence, the correct option is (b).

6. Oxygen is collected by downward displacement of water, as it has negligible solubility in water.

Hence, the correct option is (a).

7. Since the ignition temperature of LPG is less, it can be lighted by automatic gas lighter. Kerosene stone cannot be lighted by automatic gas lighter since kerosene has a higher ignition temperature and the spark produced by the lighter cannot rise the temperature of the fuel to that extent.

Hence, the correct option is (c).

8. Tinning is used to prevent corrosion of containers that are used for storing edible materials.

Hence, the correct option is (d).

9. The major source of most of the air pollutants is automobiles.

Hence, the correct option is (b).

10. Oxyhydrogen flame is used for welding.

Hence, the correct option is (d).

11. Thermal decomposition of Ag_2O gives Ag; hence, silvery white globules are formed.

Hence, the correct option is (b).

12. The metal used is Pt.

Hence, the correct option is (b).

13. Oxygen gas turns alkaline pyrogallol to dark brown.

Hence, the correct option is (a).

14. The innermost zone is also known as zone of unburnt wax vapours, due to the presence of unburnt carbon particles.

Hence, the correct option is (c).

15. Safe disposal of industrial waste is a measure to control acid rain but not global warming.

Hence, the correct option is (b).

Assessment Test IV

1. Mercury is a liquid with high density. Hence, it can rise as a short column when the pressure exactly balances the atmospheric pressure. Due to weak adhesive force between Hg and glass, it does not stick to the wall of the glass tube.

Hence, the correct option is (c).

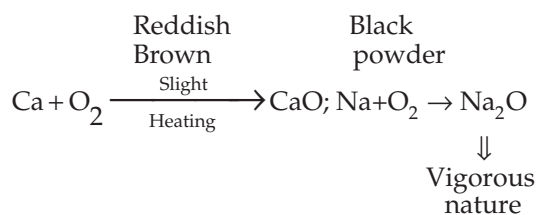
2. In the preparation of O_2 from H_2O_2 , no heating is required.

Hence, the correct option is (b).

3. The correct order is BACD.

Hence, the correct option is (a).

4. $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_3\text{O}_4$; $\text{Cu} + \text{O}_2 \rightarrow \text{Cu}_2\text{O}$



Hence, the correct option is (a).

5. (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (A)

Hence, the correct option is (b).

6. Oxygen is slightly soluble in water.

Hence, the correct option is (c).

7. Among the given fuels, CNG has the least ignition temperature.

Hence, the correct option is (c).

8. Surgical instruments, utensils, etc. are prevented from rusting by alloying.

Hence, the correct option is (a).

9. CFC and fly ash are not produced by automobiles.
Hence, the correct option is (c).
10. In photosynthesis, O_2 is not used.
Hence, the correct option is (c).
11. $2PbO_2 \rightarrow 2PbO + O_2$
Brown Yellow
Hence, the correct option is (c).
12. $SO_2 + O_2 \xrightarrow{V_2O_5 \text{ or } Pt} SO_3$.
(X) (Y)
Hence, the correct option is (a).
13. $NO + O_2 \rightarrow NO_2$
Colourless Reddish brown
(X) (Y)
Hence, the correct option is (c).
14. Luminous zone produces soot due to incomplete combustion.
Hence, the correct option is (d).
15. Corrosion of monuments takes place due to acid rain.
Hence, the correct option is (a).

CHAPTER 5

Water, Solution, Solubility and Hydrogen

Assessment Test I

1. (A) 100 g ice $\rightarrow$ water at 0°C

$$q_1 = 100 \times 80 = 8000 \text{ cal}$$

Water at $0^\circ\text{C} \rightarrow$ water at 30°C

$$q_2 = 100 \times 1 \times 30 = 3000 \text{ cal}$$

$$Q = q_1 + q_2 = 8000 + 3000 = 11000 \text{ cal}$$

- (B) 50 g boiling water $\rightarrow$ water vapour at 100°C

$$q_1 = 50 \times 540 = 27,000 \text{ cal}$$

Water vapour at $100^\circ\text{C} \rightarrow$ water vapour at 120°C

$$q_2 = 50 \times 0.9 \times 20 = 900 \text{ cal}$$

$$Q = q_1 + q_2 = 27,000 + 900 = 27,900 \text{ cal}$$

- (C) 100 g boiling water $\rightarrow$ water vapour at 100°C

$$q = 100 \times 540 = 54,000 \text{ cal}$$

- (D) 50 g ice $\rightarrow$ water at 0°C

$$q = 50 \times 80 = 4000 \text{ cal}$$

Increasing order of heat required is DABC.

Hence, the correct option is (b).

2. Due to the high value of dielectric constant of water, water can break the forces of attraction between the oppositely charged ions. Therefore, many ionic compounds are highly soluble in water.

Hence, the correct option is (a).

3. The formulae of blue vitriol and washing soda are $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$, respectively. Both of them possess water of crystallization.

Hence, the correct option is (c).

4. At 40°C , the order of solubility of substances according to the solubility curve is $B < A < D < C < E$.

Hence, the correct option is (a).

5. Solubility = 40

Mass of saturated solution = 210 g

140 g of solution $\rightarrow$ 40 g of substance

$$210 \text{ g of solution} \rightarrow \frac{40 \times 210}{140} = 60 \text{ g}$$

Hence, the correct option is (c).

6. Solubility of calcium sulphate first increases and then decreases with temperature. Therefore, it shows the same solubility at two different temperatures.

Hence, the correct option is (b).

7. Moisture catalyses the reaction between hydrogen and chlorine to form hydrogen chloride. Therefore, no reaction takes place between dry hydrogen and chlorine even in the presence of diffuse or intense sunlight

Hence, the correct option is (a).

8. Solubility of a gas in a liquid is proportional to the external pressure and inversely proportional to the external temperature. As atmospheric pressure is the least at hill stations, solubility is also the least. The solubility is greater in winter than in summer. The solubility of oxygen in water is maximum in deep sea in winter. The correct order is deep sea in winter $>$ sea level in winter $>$ sea level in summer $>$ hill station in summer.

Hence, the correct option is (b).

9. 1 g of water requires 4.2 J of heat for rise in the temperature by 1°C . 10 g of water requires 42 J of heat for the same rise in temperature. For reaching the boiling point (100°C), 4200 J or 4.2 kJ of heat is required.

Hence, the correct option is (c).

10. The presence of dissolved salts decreases the freezing point of water. Distilled water freezes at a higher temperature than saline water. Increase in pressure decreases the freezing point. As pressure is low at higher altitudes, the water freezes at a higher temperature at hill station than on plains.

Hence, the correct option is (b).

11. (i) $\rightarrow$ (C), $K + H_2O \rightarrow KOH + H_2 \Rightarrow$ Catches fire with lilac flame
 (ii) $\rightarrow$ (D), $Na + H_2O \rightarrow NaOH + H_2 \Rightarrow$ Catches fire with yellow flame
 (iii) $\rightarrow$ (E); $Mg + H_2O \rightarrow Mg(OH)_2 + H_2 \Rightarrow$ Brilliant white light and white ash
 (iv) $\rightarrow$ (A); $Zn + H_2O \rightarrow ZnO + H_2 \Rightarrow$ Yellow substance and turns to white on cooling
 (v) $\rightarrow$ (B); $Fe + H_2O \rightarrow Fe_2O_3 + H_2 \Rightarrow$ Brown substance

Hence, the correct option is (c).

12. Lead is not a preferred metal for the laboratory preparation of hydrogen because of the formation of insoluble $PbCl_2$ or $PbSO_4$, which stops the further reaction.

Hence, the correct option is (b).

13. When a glass full of ice is subjected to melting, the water level in the glass will be lower than the ice. This is because ice has lower density than water and occupies greater volume than water. The water formed is denser than ice occupies lesser volume.

Hence, the correct option is (c).

14. X is a saturated solution as the solubility of the substance at 80°C is 80. Y is an unsaturated solution as the solubility of the substance at 80°C is 50 and the solution contains only 50 g of substance in it. Since X is a saturated solution and its solubility decreases with temperature, X starts crystallising on cooling just below 80°C itself. However, Y becomes a saturated solution at 50°C where the solubility of the substance is 50 and it starts crystallizing just below 50°C .

Hence, the correct option is (c).

15. At 0°C , the solubilities of X and Y are 20 and 10, respectively. That means, 20 g of X and 10 g of Y will be left behind in the solution even at 0°C . Therefore, out of 50 g of the substances in the solution taken at 80°C , 30 g of X and 40 g of Y can be precipitated.

Hence, the correct option is (a).

Assessment Test II

1. (A) 100 g water at $50^\circ\text{C} \rightarrow$ water at 0°C

$$q_1 = 100 \times 1 \times 50 = 5000 \text{ cal}$$

Water at $0^\circ\text{C} \rightarrow$ ice at 0°C

$$q_2 = 100 \times 80 = 8000 \text{ cal}$$

$$Q = q_1 + q_2 = 5000 + 8000 = 13,000 \text{ cal}$$

- (B) 50 g water at $0^\circ\text{C} \rightarrow$ ice

$$q = 50 \times 80 = 4000 \text{ cal}$$

- (C) 100 g water vapour $\rightarrow$ water at 100°C

$$Q = 100 \times 540 = 54,000 \text{ cal}$$

- (D) 50 g water vapour $\rightarrow$ water at 100°C

$$q_1 = 50 \times 540 = 27000 \text{ cal}$$

Water at $100^\circ\text{C} \rightarrow$ water at 50°C

$$q_2 = 50 \times 1 \times 50 = 2500 \text{ cal}$$

$$Q = q_1 + q_2 = 27000 + 2500 = 29,500 \text{ cal}$$

Decreasing order of heat evolved is CDAB.

Hence, the correct option is (d).

2. Glucose is not an inorganic compound. Its high solubility in water is not due to high dielectric constant.

Hence, the correct option is (b).

3. Glauber's salt and washing soda are efflorescent substances. On exposure to air, Glauber's salt loses water of crystallization completely and forms an anhydrous salt. However, washing soda loses only 9 water molecules and forms a monohydrate. Only on heating, it forms anhydrous salt.

Hence, the correct option is (c).

4. The decreasing order of solubility of given substances at the given temperatures is ECDBA.

Hence, the correct option is (c).

5. Mass of the solution = 80 g

Mass of solute = 16 g

Mass of solvent = 80 – 16 = 64 g

Amount of substance dissolved in 64 g water = 16 g

Amount of substance dissolved in 100 g = $\frac{1600}{64} = 25$ g

Solubility of the substance = 25

Hence, the correct option is (d).

6. Glauber's salt is $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$. Its solubility increases with increase in temperature. At 32.6°C , it changes to anhydrous Na_2SO_4 . Its solubility decreases with increase in temperature. Change in water of crystallization is not the reason for change in the trend of solubility.

Hence, the correct option is (b).

7. Red phosphorus burns in oxygen in the presence of moisture which catalyses the reaction. Anhydrous CaCl_2 absorbs moisture from the air in contact with phosphorus. Therefore, no reaction occurs in the test tube.

Hence, the correct option is (d).

8. Collection of hydrogen by downward displacement of water is due to the insoluble nature of hydrogen in water.

Hence, the correct option is (d).

9. 4.2 J of heat is required for 1 g of water for 1°C rise in temperature. 100 g of water requires 420 J of heat.

420 J of heat $\rightarrow 1^\circ\text{C}$

$2.1 \rightarrow 10^3$ J of heat $\rightarrow \frac{2.1 \times 10^3}{420} = 5^\circ\text{C}$

Hence, the correct option is (d).

10. As brine solution is a saturated solution of sodium chloride, it contains maximum amount of dissolved salt and hence, freezes at the lowest temperature.

Hence, the correct option is (a).

11. (i) $\rightarrow$ (C); Reaction of potassium with water is vigorous and exothermic.

(ii) $\rightarrow$ (D); The reaction becomes reversible when carried out in a closed container.

(iii) $\rightarrow$ (B); A milky alkaline solution of $\text{Ca}(\text{OH})_2$ is formed by the reaction of calcium with water.

(iv) $\rightarrow$ (A); Magnesium reacts only with boiling water.

Hence, the correct option is (d).

12. When iron is taken in place of zinc for the laboratory preparation of hydrogen, the reaction becomes slow and yields only lesser volume of hydrogen as the reaction of iron with acid becomes reversible.

Hence, the correct option is (d).

13. Floating of dead body in the water in the Dead Sea is due to the greater density of water in dead sea due to the very high salinity of that water. This is not a consequence of anomalous expansion of water.

Hence, the correct option is (a).

14. At 35°C , X and Y possess same solubility.

Hence, the correct option is (a).

15. The solution of X and Y contain 90 g and 70 g of solutes at 80°C as they are saturated solutions. When they are cooled up to 60°C , 20 g of X and 10 g of Y are precipitated since their solubilities are 70 and 50 at 60°C .

Hence, the correct option is (b).

Assessment Test III

1. During the formation of snow, water vapour directly gets converted to ice.

Hence, the correct option is (d).

2. Density of water is less than 1 g/cc above (or) below 4°C . Hence, the density of water at 2°C could be < 0.95 g/cc

Hence, the correct option is (a).

3. Distilled water is a bad conductor of electricity since it is devoid of any dissolved salts.

Hence, the correct option is (d).

4. 1 g of steam releases 540 calories heat.

80 calories of heat are required to melt 1 g ice at 0°C to 1 g water at 0°C .

$$\therefore x = \frac{540}{80} = 6.75 \text{ g}$$

Hence, the correct option is (a).

5. Heat released = $m \times s \times \Delta t$

$$= 10 \times 4.2 \times (50 - 30)$$

$$= 10 \times 4.2 \times 20$$

$$= 840 \text{ J}$$

Hence, the correct option is (a).

6. Water can dissolve many inorganic salts in it because it has high dielectric constant.

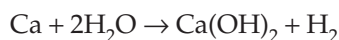
Hence, the correct option is (c).

7. SO_2 is an acidic oxide since it forms H_2SO_3 on reaction with water.

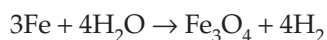
Hence, the correct option is (d).

8. $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$

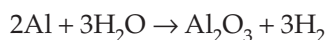
1 molecule Na liberates 0.5 molecules of H_2 .



1 molecule of Ca liberates 1 molecule of H_2 .



1 molecule of Fe liberates $\frac{4}{3} = 1.33$ molecule of H_2 .



1 molecule of Al gives rise to $\frac{3}{2} = 1.5$ molecule of H_2

$\therefore$ The order is ABDC.

Hence, the correct option is (a).

9. The order is DABC.

Hence, the correct option is (c).

10. 25 g of the salt is present in $225 - 25 = 200$ g of the solvent.

$$\therefore \text{Solubility} = \frac{25}{200} \times 100 = 12.5$$

Hence, the correct option is (b).

11. Since the solubility of B is maximum at 40°C , it is present in the maximum amount in its saturated solution.

Hence, the correct option is (b).

12. Efflorescent salts release water at room temperature.

Hence, the correct option is (b).

13. Solubility of a gas increases with the increase in pressure.

Hence, the correct option is (b).

14. Water gas is $\text{CO} + \text{H}_2$

Hence, the correct option is (c).

15. The correct matching is

(i) $\rightarrow$ (D)

(ii) $\rightarrow$ (E)

(iii) $\rightarrow$ (B)

(iv) $\rightarrow$ (A)

(v) $\rightarrow$ (C)

(vi) $\rightarrow$ (F)

Hence, the correct option is (d).

Assessment Test IV

1. Water vapour gets converted to water during the formation of dew.

Hence, the correct option is (a).

2. Ice occupies slightly more volume than water at 0°C since density of ice is lesser than the water.

Hence, the correct option is (b).

3. The electrical conductivity of water is due to the presence of dissolved salts.

Hence, the correct option is (a).

4. Let x calories is the amount of heat supplied.

$$\frac{\text{Mass of ice}}{\text{Mass of water}} = \frac{\frac{x}{80}}{\frac{x}{540}}$$

$$= \frac{540}{80} = 27:4$$

Hence, the correct option is (b).

5. Heat supplied = $m \times s \times \Delta t$

$$= 5 \times 4.2 \times (75 - 25)$$

$$= 5 \times 4.2 \times 50$$

$$= 1050 \text{ J}$$

Hence, the correct option is (b).

6. Different inorganic salts dissolve in water to a different extent. Dielectric constant of water does not change with the change of the solute.

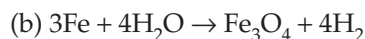
Hence, the correct option is (c).

7. Metallic oxides are basic oxides, and hence, they react with acid. They form bases when they react with water.

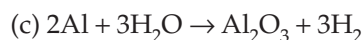
Hence, the correct option is (d).

8. (a) $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$

1 molecule of K $\rightarrow$ 1 molecule of H_2O



1 molecule of Fe $\rightarrow \frac{4}{3} = 1.33$ molecule of H_2O



1 molecule of Al $\rightarrow \frac{3}{2} = 1.5$ molecule of Fe

$\therefore$ The expected order is c b a.

Hence, the correct option is (d).

9. The order is BAC.

Hence, the correct option is (b).

10. $30 = \frac{x}{(390 - x)} \times 100$, x is the amount of solute (B).

$$\therefore 390 \times 30 - 30x = 100x$$

$$\therefore 130x = 390 \times 30$$

$$\therefore x = \frac{390 \times 30}{130} = 90 \text{ g}$$

Hence, the correct option is (a).

11. 100 g water contains $\frac{150}{750} \times 100$.

= 20 g of the salt

N is unsaturated.

Hence, the correct option is (c).

12. Hydrated calcium chloride is a deliquescent salt.

Hence, the correct option is (d).

13. Solubility of a gas is inversely proportional to the temperature. Hence, boiling expels the dissolved gases.

Hence, the correct option is (b).

14. $\text{CO} + \text{H}_2 + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 2\text{H}_2$

CO is oxidized to CO_2 .

Hence, the correct option is (a).

15. The correct matching is:

(i) $\rightarrow$ (F); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (C); (v) $\rightarrow$ (D); (vi) $\rightarrow$ (E)

Hence, the correct option is (a).

CHAPTER 6

Carbon and Its Compounds

Assessment Test I

1. Diamond is a good thermal conductor because of the compact three-dimensional structure. Mere tetravalency of carbon does not make it a good thermal conductor.

Hence, the correct option is (b).

2. Dry ice undergoes sublimation by taking heat from the surroundings, thereby causing cooling, which results in the condensation of water vapour and ultimately results in rain.

Hence, the correct option is (a).

3. (i) $\rightarrow$ (C)
(ii) $\rightarrow$ (D)
(iii) $\rightarrow$ (E)
(iv) $\rightarrow$ (A)
(v) $\rightarrow$ (B)

Hence, the correct option is (c).

4. (i) Extraction of yellow phosphorus $\rightarrow$ Bone charcoal
(ii) Preparation of artificial diamonds $\rightarrow$ Sugar charcoal
(iii) Gas masks $\rightarrow$ Wood charcoal
(iv) Moderator in nuclear reactors $\rightarrow$ Graphite
(v) Abrasive for cutting glass $\rightarrow$ Diamond

Hence, the correct option is (b).

5. (i) Passing CO gas over finely divided iron $\rightarrow$ $\text{M}(\text{CO})_5$ or $\text{Fe}(\text{CO})_5$
(ii) Passing CO_2 gas through powdered magnesium $\rightarrow$ MO or MgO
(iii) Heating M_2O_3 with $\text{CO} \rightarrow \text{M}$ or Fe
(iv) Passing CO_2 through sodium $\rightarrow$ M_2CO_3 or Na_2CO_3

Hence, the correct option is (c).

6. Except sodium and potassium carbonates, all other carbonates, on heating, decompose to produce CO_2 .

Hence, the correct option is (a).

7. Baking powder contains sodium bicarbonate as a major component, which, on heating, decomposes to release CO_2 .

Hence, the correct option is (b).

8. A liquid CO_2 fire extinguisher works on the principle of evaporation of liquid CO_2 to gaseous state, thereby reducing the temperature of the combustible substance.

Hence, the correct option is (d).

9. Usage of graphite as a material of refractory crucibles is due to the high melting point and thermal resistance of graphite and not due to its layer lattice structure.

Hence, the correct option is (a).

10. Reaction of carbonic acid with NaOH results in the formation of Na_2CO_3 which is highly soluble in water. Therefore, it does not involve the formation of white precipitate, whereas all other reactions result in the formation of insoluble carbonates, which are obtained as white precipitates.

Hence, the correct option is (a).

11. Coal, on destructive distillation, gives various products. X should be coke since it can reduce iron oxide to iron, whereas Y should be coal tar since it gives benzene as the final product on fractional distillation.

Hence, the correct option is (c).

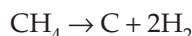
12. The general formula corresponding to saturated hydrocarbons should be C_nH_{2n+2} . According to this, the formula corresponding to the one with 4 carbon atoms should be C_4H_{10} .

Hence, the correct option is (c).

13. Marsh gas (methane) is the harmful gas released in coal mines.

Hence, the correct option is (d).

14. Methane on pyrolysis gives carbon black.



Hence, the correct option is (a).

15. High ignition temperature is not the characteristic of an ideal fuel. It should always have low ignition temperature.

Hence, the correct option is (c).

Assessment Test II

1. Graphite is a good electrical conductor because carbon shows a valency of three, leaving one free electron on each carbon.

Hence, the correct option is (a).

2. A sealed bottle with dry ice explodes after some time as it undergoes sublimation at room temperature and the resultant pressure developed due to the accumulation of gaseous CO_2 in the bottle. Dry ice producing low temperature has no relevance to the above.

Hence, the correct option is (b).

3. (i) $\rightarrow$ (E)
(ii) $\rightarrow$ (D)
(iii) $\rightarrow$ (A)
(iv) $\rightarrow$ (B)
(v) $\rightarrow$ (C)

Hence, the correct option is (d).

4. (i) Sugar, molten iron $\rightarrow$ Diamond
(ii) Coke and $SiO_2 \rightarrow$ Graphite
(iii) Graphite $\rightarrow$ Fullerene
(iv) Coal $\rightarrow$ Coke

(v) Mustard oil $\rightarrow$ Lampblack

Hence, the correct option is (a).

5. (i) Neutralisation of CO_2 by caustic soda $\rightarrow$ Carbonate of sodium
(ii) Combination of CO_2 and water $\rightarrow$ Carbohydrate
(iii) $CO_2 + Cl_2 \rightarrow COCl_2$ (Phosgene)
(iv) $C + CO_2 \rightarrow CO$ (Lower oxide of carbon)
(v) CO is passed through blood $\rightarrow$ Carboxy haemoglobin

Hence, the correct option is (b).

6. X does not decompose on heating, whereas Y decomposes to give CO_2 . Among the given options, X and Y should be K_2CO_3 and $ZnCO_3$.

Hence, the correct option is (c).

7. Due to the sparingly soluble nature of CO_2 in water, the little amount of CO_2 present dissolved in water is utilised for photosynthesis by aquatic plants.

Hence, the correct option is (d).

8. Both soda-acid fire extinguisher and foam-type fire extinguisher involve a neutralization reaction between sodium bicarbonate and H_2SO_4 or $Al_2(SO_4)_3$. However, foam-type fire extinguisher can be used for extinguishing petrol or oil fires due to the presence of saponin which traps the oil vapour in the froth.

Hence, the correct option is (b).

9. Usage of graphite as a lubricant in machines, especially where high temperature is produced, involves both its layer lattice structure and high thermal resistance.

Hence, the correct option is (b).

10. Action of HCl on magnesium bicarbonate results in the formation of insoluble $MgCO_3$ which is obtained as a precipitate along with the release of CO_2 as gaseous product.

Hence, the correct option is (b).

11. Coal gas is a mixture of CO, CH_4 , and H_2 . Hence, X should be methane. Coal tar, on fractional